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11 September 1978
SUPERSEDING
MIL-I-20564D
15 March 1971

MILITARY SPECIFICATION

ICE CREAM PLANTS, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of portable ice cream plants with 2-1/2 gallon (0.01 m^3) freezer and 40 gallon (0.15 m^3) hardening capacity.

1.2 Classification. Ice cream plants covered by this specification shall be of the following types as specified (see 6.2).

Type I	-	Electrically operated.
Type II	-	Electrically operated, with auxiliary power unit consisting of gasoline engine driven generator set.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent
BB-F-1421	-	Fluorocarbon Refrigerants
CC-M-1807	-	Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
HH-I-524	-	Insulation Board, Thermal (Polystyrene)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-A-200/8 - Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, and Structural, Shapes, 6061
QQ-A-250/2 - Aluminum Alloy 3003, Plate and Sheet
QQ-A-250/8 - Aluminum Alloy, 5052, Plate and Sheet
QQ-A-596 - Aluminum Alloy, Permanent and Semipermanent Mold Castings
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy: Flux Cored Ribbon and Wire, and Solid Form
TT-E-529 - Enamel, Alkyd, Semigloss
TT-P-1757 - Primer Coating, Low Moisture Sensitivity
VV-L-825 - Lubricating Oil, Refrigerant Compressor
WW-T-799 - Tube, Copper, Seamless, (For Use with Solder-Type or Flared Tube Fittings)

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant, For the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-P-514 - Plates Identification, Instruction and Marking Blank
MIL-I-705 - Ice Cream Mix
MIL-R-1150 - Rivets, Solid (Aluminum Alloy), and Aluminum Alloy Rivet Wire and Rod
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing Of
MIL-S-12514 - Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C- Induction, 3/4 to 100 HP, Manual and Electric
MIL-R-14328 - Rubber Sheet, Synthetic, Medium Soft, General Purpose, Gasket Material (For Extreme Climatic Conditions)
MIL-G-52732 - Generator Set, Gasoline Engine Driven, 0.5 KW thru 10 KW, 60 Hertz, 400 Hertz, and 28 Volt Direct Current, Type 1 (Tactical) Class 2 (Utility)

STANDARDS

FEDERAL

FED-STD-595 - Colors

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MIL-STD-105 - Sampling for Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-279 - Compressors, Reciprocating, Power-Driven, Open-Type, for Use With Refrigerant-12
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-759 - Condensers, Air Cooled, Refrigerant-12
MS 15005 - Fittings, Lubrication, Throat or Surface Check, 1/8 Pipe Threads, Nickel - Copper Alloy Type V
MS 16993 - Plugs, Fusible, Refrigerant-12
MS 17241 - Heat Interchanger, Refrigerant-12
MS 17243 - Valve, Angle, Shut Off, Packed, Receiver, Refrigerant-12
MS 17248 - Valves, Compressor Service, Refrigerant-12
MS 17291 - Receivers, Liquid Refrigerant 12
MS 17798 - Indicator, Sight, Liquid, Refrigerant
MS 17841 - Valves, Solenoid, Refrigerant-12
MS 17843 - Controls, Pressure, Dual, Refrigerant-12; with High Pressure Cutout
MS 17982 - Valves, Expansion, Thermostatic, Refrigerant 12
MS 27211 - Thermometer, Indicating, Capillary Tube and Bulb
MS 27900 - Gages, Pressure and Gages Compound, Pressure and Vacuum, Dial Indicating, Panel Mounted, 2-1/2 Inch Dial Size, Refrigerant 12 and 22
MS 35647 - Padlocks, Key Operated
MS 35845 - Dehydrators, Desiccant, Refrigerant-12
MS 35867 - Connection End, Internal Flare, Refrigeration
MS 35868 - Connection End, External Pipe Thread; Refrigeration
MS 35869 - Adapter, Straight, Pipe to Tube (Flare to External Pipe)
MS 35871 - Tees, Pipe to Tube, Refrigeration
MS 35872 - Flare Nut and Flare Seal Bonnet, Refrigeration
MS 35873 - Connection, End, External Flare, Refrigeration
MS 35880 - Valves, Packless, Stop, Solder-Joint Connection Ends
MS 35881 - Valves, Packless, Stop, Flare Connection Ends
MS 35882 - Valves, Packless, Stop, Three-Way, Solder-Joint Connection Ends
MS 35883 - Valves, Packless, Stop, Three-Way, Flare Connection Ends
MS 35917 - Elbows, Tube, 90 and 45 Degree, Solder-Joint, Refrigeration

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MS 35918 - Adapter, Straight, Pipe Thread to Tube, Male Solder-Joint, Refrigeration
MS 35919 - Adapter, Straight, Pipe to Tube, External Flare to Female Solder-Joint, Refrigeration
MS 35920 - Reducer, Tube, Solder-Joint, Refrigeration
MS 35921 - Adapter, Straight, Pipe to Tube, Pipe Thread to Female Solder-Joint, Refrigeration
MS 35922 - Elbow, Tube, 90 Degree, Reducing, Solder-Joint Refrigeration
MS 35923 - Reducer, Tube, Solder-Joint, Refrigeration
MS 35924 - Plug and Cap, Tube, Solder-Joint, Refrigeration
MS 35925 - Connection End, Solder-Joint, Refrigeration
MS 35926 - Connection End, Internal Pipe Thread, Refrigeration
MS 35927 - Coupling, Tube, with Stop, Solder-Joint, Refrigeration
MS 35928 - Elbows, Tube, Street, 90 and 45 Degree, Solder-Joint Refrigeration
MS 35929 - Tee, Tube, Solder-Joint, Refrigeration

DRAWINGS

ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-212 - Assembly
5-13-213 - Exterior Views and Details
5-13-214 - Sectional Plan
5-13-215 - Cross Section
5-13-216 - Longitudinal Section
5-13-217 - Frame Assembly
5-13-218 - Frame Assembly and Details
5-13-284 - Engine-Generator Frame Assembly

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1692 - Flammability of Plastics, Foams and Sheeting

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

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INTERNATIONAL ASSOCIATION OF MILK AND FOOD SANITARIANS, INC.

3-A Sanitary Standards for Batch and Continuous Freezers

for Ice Cream, Ices and Similarly Frozen Dairy Foods

(Application for copies should be addressed to International Association of Milk and Food Sanitarians, Inc., Box 437, Shelbyville, IN 46176.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS

(ASHRAE)

Standard No. 15 - Safety Code for Mechanical Refrigeration

(Application for copies should be addressed to American Society of Heating, Refrigerating and Air Conditioning Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

UNITED STATES DEPARTMENT OF COMMERCE

PS-1 - Softwood Plywood - Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.2, 6.2, and 6.3).

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3.2 Standards.

3.2.1 Safety standard. Unless otherwise specified herein, the ice cream plants shall conform to the American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) Standard No. 15.

3.2.2 Sanitary standard. The ice cream plants shall conform to the requirements of 3-A Sanitary Standards indicated in 2.2 and requirements as specified hereinafter.

3.2.3 Compliance. Prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the ice cream plants he proposes to furnish under this specification meet the requirements of ASHRAE Standard No. 15 and the 3-A Standard for Batch and Continuous Freezers for Ice Cream, Ices and Similarly Frozen Dairy Foods. Acceptable evidence of meeting the requirements of these standards shall be a certified statement by an executive of the manufacturer or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the ice cream plants offered have been tested and conform to the safety standard and the sanitary standard.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for ice cream plants, provided the completed end item complies with all provisions of this specification. Unless otherwise specified herein, commercial tolerances are applicable to material thicknesses and diameters. All ferrous components not specifically specified herein shall be either hot-dipped galvanized or electro-galvanized (see 6.7).

3.3.1 Tube, copper. Tube used for refrigerant lines with an outside diameter of 5/8 inch (16 mm) or less shall be soft copper tube, cleaned and dehydrated, normally used by the refrigeration industry. Refrigerant line tube used with an outside diameter of 7/8 inch (22 mm) or larger, shall be type K or L of WW-T-799. Wall thickness of tube to be used shall be as follows:

1/2 inch (13 mm) outside diameter (OD)	- 0.032 inch (0.8 mm) wall thickness
5/8 inch (16 mm) OD	- 0.035 inch (0.9 mm) wall thickness
7/8 inch (22 mm) OD	- Wall thickness per WW-T-799

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3.3.2 Valves, fittings, and connections. Valves shall conform to MS 35880 through MS 35883. Fittings shall conform to MS 15005, MS 35871 through MS 35873, MS 35917 through 35929. The solder-joint and flared connections shall conform to MS 35867 through MS 35869.

3.3.3 Bolts, nuts, screws, and washers. Bolts, nuts, screws, and washers used for mounting components shall be of the quality and in the sizes, types, and materials normally used for the purpose in commercial practice. Threads shall conform to the National Bureau of Standards Handbook H28. Steel bolts, nuts, screws, and washers for the evaporator shall be zinc plated. Brass bolts, nuts, washers, and machine screws shall not be used in fastening aluminum parts.

3.3.4 Solder. Solder shall conform to composition Sb5 of QQ-S-571.

3.3.5 Insulation. All insulation shall be expanded polyurethane or polystyrene and shall have "K" factors specified below using a 75°F (24°C) mean temperature, expressed in British thermal units (Btu) per hour per square foot per degree F. temperature difference per inch of thickness (W/M·K). Loose fill type insulation shall not be acceptable.

3.3.5.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to class 2, type I or II of HH-I-524.

3.3.5.2 Foam-in-place. Foam-in-place insulation shall be self extinguishing polyurethane material and shall have a manufacturer's rated "K" factor of not more than 0.15. The insulation shall be fire retardant and self extinguishing as determined by ASTM D 1692. To meet the relative burning characteristics of this specification, the words fire retardant and self extinguishing shall be construed to mean that the flame shall extinguish within 60 seconds after removal of the burner. Further, no insulation sample shall burn for a distance greater than 2-1/4 inches (57 mm) from the end exposed to the test flame. No flaming dripping particles or droplets shall emit from the samples. The insulation shall have a free rise density of not less than 1.7 pounds per cubic foot (27.2 kg/m^3) when tested as specified in 4.4.9.

3.3.6 Lumber. All lumber shall be select grade hardwood.

3.3.6.1 Plywood. All plywood shall be group 1, exterior type, grade B-C or better, sanded on two sides and shall conform to PS-1.

3.3.6.2 Preservative, wood. Wood preservative shall be a solution conforming to P-D-680 and containing one percent solubilized copper-8-quinolinolate. Wood shall be immersed for a period of not less than 15 minutes and air dried until free of solvent. The solubilized copper-8-quinolinolate shall be free of amines and naphthenic acid or its derivative.

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3.3.7 Gaskets. Gaskets used on lids of the hardening cabinet shall be of the cushion-type, with a synthetic or natural rubber core incased in a synthetic rubber covering. Synthetic rubber shall conform to MIL-R-14328.

3.3.8 Metal surface treatment. Aluminum surfaces to be painted shall be given a preparatory treatment conforming to class 2 of MIL-C-5541.

3.3.9 Metal surface primers and paints.

3.3.9.1 Primer, zinc chromate. The zinc chromate primer shall conform to TT-P-1757.

3.3.9.2 Finish coat. Unless otherwise specified (see 6.2), the finish coat on aluminum surfaces shall be a semigloss paint conforming to TT-E-529, color number 24087 of FED-STD-595.

3.3.10 Refrigerant. The refrigerant shall conform to type 12 of BB-F-1421.

3.3.11 Oil, lubricating, compressor. Refrigerant compressor lubricating oil shall conform to type II of VV-L-825 or may be oil recommended by compressor manufacturer.

3.3.12 Aluminum.

3.3.12.1 Sheet or plate. The aluminum sheet or plate shall conform to QQ-A-250/8, or QQ-A-250/2.

3.3.12.2 Castings. The aluminum castings shall be sand or permanent mold fabricated. When sand castings are supplied, they shall conform to alloy numbers B195, 355, or 356 of QQ-A-596.

3.3.12.3 Bar, rod, and shapes. Aluminum bar, rod, and shapes shall conform to QQ-A-200/8.

3.3.12.4 Rivets. Aluminum rivets shall conform to MIL-R-1150.

3.4 Design. The ice cream plants shall be capable of producing 2-1/2 gallons (0.01 m^3) of ice cream per batch and hardening 40 gallons (0.15 m^3) of ice cream in the hardening cabinet. The ice cream plants shall have an electrically powered refrigeration system. All components shall be integrally mounted within a structural frame and unless otherwise specified (see 6.2) the plant shall be designed for operation from a nominal 208-volt, 4-wire, 60-Hertz (Hz), 3-phase external power source. The type II units shall be identical to the type I units, except the type II units shall be furnished with an auxilary power source consisting of a gasoline engine driven generator set. Provision shall be made in the framework housing to allow for manual cranking of the engine without removing any panels. The ice cream plants shall be portable and free standing when assembled and all components shall operate as intended when tested as specified in 4.4.5.

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3.4.1 Capacity. The freezer section of the ice cream plant shall deliver 2-1/2 gallon (0.01 m^3) batches of ice cream within 15 minutes with a minimum of 95 percent overrun and a maximum of 110 percent at 23° Fahrenheit (F) (-5°C), ± 1°F (± 0.5°C) using a 90°F (32°C) mix conforming to MIL-I-705 and operating in 110°F (43°C) ambient. During 6 hours of continuous operation, the freezer section produces at least 40 gallons (0.15 m^3) of ice cream. The hardening cabinet shall reduce 40 gallons (0.15 m^3) of semifrozen ice cream as it comes from the freezer to 5°F (-15°C) or lower in 18 hours of operation (see 4.4.6).

3.5 Construction. The ice cream plants shall be constructed and assembled as shown on Drawings 5-13-212 thru 5-13-218 and 5-13-284. All component parts of the ice cream plants shall be mounted to minimize vibration. All accessories, instruments, controls, and tubing shall be fastened or braced to prevent damage or loosening during operation or transit. All refrigeration components shall be gas tight when tested as specified in 4.4.3. The inner liner of the hardening cabinet shall be watertight when tested as specified in 4.4.8. All electrical equipment shall be protected from the damaging effect of rainfall. Electrical components having a temperature rise of not more than 40° Centigrade (C) shall be coated with mildew resistant varnish in accordance with MIL-V-173. All wood members used in the fabrication of the ice cream plants shall have a preservative treatment in accordance with 3.3.6.2. Necessary trimming or resurfacing of completely milled, treated, and kiln-dried wood members shall be permitted to secure the required fit provided surfaces are thoroughly coated prior to assembly with wood preservative specified in 3.3.6.2. All gaskets shall be made from the material specified in 3.3.7 and shall be close fitting and symetrically installed to provide airtight sealing (see 4.4.4). Contact areas between dissimilar metals shall be insulated to minimize galvanic corrosion by coating the surface of both metals with zinc chromate or asphaltic type compounds or by strips of rubber conforming to MIL-E-14328. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without removing any major component. Access panels, covers, and plates shall be provided as required for component adjustment, repair, or replacement and shall be fastened with easily operated, quick-disconnect fasteners or with screws. The assembled ice cream plant shall not be visibly damaged as a result of the drop test specified in 4.4.1.

3.5.1 Insulation installation. All space within the walls of the hardening cabinet, including the lids and freezer head coil shall be completely filled with pre-expanded insulation specified in 3.3.5.1 or the foam insulation specified in 3.3.5.2. The insulation shall be installed to leave no voids or empty pockets. Pre-expanded insulation when used, and if installed in layers, shall have the layers bonded to one another with a minimum of joints, carefully filled and staggered. No layer shall be less than 2 inches (51 mm) thick. Pre-expanded insulation shall be bonded to the inner surfaces of the inner liner and outer shell by a commercial grade permanent cement. When copper tube is used for the evaporator, voids between the tubes shall be filled with a thermosetting mastic.

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3.5.2 Freezer head section. Leakage of lubricant into the freezing chamber or of mix from the freezing chamber shall be prevented by a ground-joint seal and an "O" ring seal where necessary.

3.5.2.1 Rear head. The rear head, if separate, shall be secured and sealed to the freezer shell. The rear head shall contain the whipper and dasher drive seal and sealed drive bearings where used and drive mechanism.

3.5.2.2 Front head. The front head shall contain a removable door and the front head shall be secured and sealed to the freezer shell.

3.5.2.2.1 Freezer head door. The freezer head door shall contain the front whipper and dasher bearings, filler spout with cover, draw-off gate, and shall be leakproof with a ground-joint seal with the stationary section of the front freezer head when in a closed position. Freezer door may have an "O" ring seal in addition to ground joint seal. The freezer door shall be held in the closed position by hand operated fasteners and shall be removable to permit sanitizing operations.

3.5.3 Hardware. Each lid shall have two malleable or die cast zinc hinges, one malleable iron or die cast zinc type handle, and one malleable iron or die cast zinc lever fastener and wedge secured to the framing. Each lever fastener shall have provision to accept a padlock conforming to MS 35647-5 of MS 35647. Hinges shall have a stop at 105 degrees from the closed position of the lid. Aluminum sheet metal shims (see 3.3.12.1) not more than 1/16 inch (1.6 mm) thick shall be placed under the base of the lever fastener and under the hinge butts in sufficient number to provide uniform compression of the gasket and effect a completely airtight seal (see 4.4.4).

3.5.4 Baskets. Twelve wire baskets shall be furnished with the ice cream plant. The overall dimensions shall be 6-1/4 inches (159 mm) wide, 20-1/8 inches (512 mm) long, and 12 inches (304 mm) deep, ± 1/4-inch (± 6 mm) on all dimensions. The basket design shall include a top frame wire having a minimum diameter of 0.135 inch (3 mm) and a rib and body wire having a minimum diameter of 0.11 inch (3 mm). The body or rib wires shall be welded to the top frame wire around the perimeter on 1-1/8 inch (28 mm) centers. A reinforcing rib wire shall be welded to each vertical wire parallel to and 10-1/2 inches (267 mm) below the top frame wire. The end body wires extending lengthwise shall be further reinforced by the side body wires extending across the basket to form the bottom, passing under them and welded at each right angle intersection. The baskets shall be "U" shaped and capable of free standing. The entire basket shall be tinned by the hot-dip or electrodeposited method after fabrication.

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3.5.5 Inner liner and evaporator. The inner liner shall be as shown on drawings. The material shall be either galvanized steel or stainless steel, with a minimum thickness of 0.0598 inch (1 mm) for galvanized steel or 0.0478 inch (1 mm) stainless steel. The evaporator may be an intergal part of the inner liner or may be copper tubing fastened to the insulation side of the inner liner. The maximum temperature difference between the hardening cabinet air temperature and the saturated refrigerant temperature at the compressor suction valve inlet shall not exceed 25°F (-4°C) (see 4.4.6) after pulldown.

3.5.6 Breaker strip. The breaker strip shall be as shown on drawings or may be fabricated from extruded rubber breaker strips of the same sections shown and installed with all joints vulcanized or sealed.

3.5.7 Refrigeration system. The refrigeration system shall consist of an electric motor driven refrigeration compressor, condenser, receiver, evaporator, controls, and all other accessories necessary to operate as a complete refrigeration system.

3.5.7.1 Thermal expansion pressure limiting valves. Thermal expansion pressure limiting valve shall be provided conforming to MS 17982.

3.5.7.2 Freezer head, liquid line solenoid valve. A liquid line solenoid valve for the freezer head shall be provided conforming to MS 17841.

3.5.7.3 Heat interchanger. A heat interchanger shall be provided conforming to MS 17241.

3.5.7.4 Condensing unit assembly. The condensing unit assembly shall be aircooled, and of sufficient capacity to meet the requirements specified in 3.4 and 3.4.1. The compressor shall be an open-type. The condensing unit shall be complete with fan of sufficient size to insure air circulation over the condenser when the ice cream plant is operated under conditions specified in 3.4 and 3.4.1. Provisions shall be made in the condensing unit compartment for drainage of any water which might collect in the bottom.

3.5.7.5 Condenser. The condenser provided shall conform to MIL-STD-759.

3.5.7.6 Fan. The fan for the condenser shall be direct motor driven conforming to MIL-STD-759. The fan motor shall be 1/6 horsepower (H.P.) (124 W) minimum, and unless otherwise specified (see 6.2), operated from a nominal 120-volt, 0 Hz 1-phase source.

3.5.7.7 Compressor. The compressor shall be provided conforming to MIL-STD-279. Vibration eliminators shall be used in the suction and discharge lines. The flexible portion of the vibration eliminators shall be a minimum of 7 inches (178 mm) long.

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3.5.7.8 Electric motor. The electric motor shall be a 3 H.P. (224 KW), 1750 revolutions per minute (rpm) NEMA design "B" with locked rotor torque 215 percent of full load torque, continuous duty, protected by a drip proof inclosure and shall conform to CC-M-1807. A thermal overload protective device shall be supplied with a rating as to open the electrical circuit before the windings exceed the temperature rise for which the motor is rated. The ambient temperature rise shall not exceed 50°C. Fuses shall not be acceptable in lieu of the thermal overload protective device.

3.5.7.9 Drive-pulleys and belts. Power shall be transmitted from the electric motor to the compressor through two V-belts. All pulleys shall be designed for operation with A or B section belts. Provision for belt adjustment shall be made by moving of electric motor by means of compression or tension bolts.

3.5.7.10 Dehydrator. The dehydrator shall conform to MS 35845. The dehydrator shall be installed with a three valve bypass including the liquid line receiver valve.

3.5.7.11 Fusible plugs. Fusible plugs shall conform to MS 16993.

3.5.7.12 Liquid line sight indicator and moisture detector. The liquid line sight indicator and moisture detector shall be provided conforming to MS 17798.

3.5.7.13 Receiver. The receiver shall be provided conforming to MS 17291, and shall include inlet and outlet valves conforming to MS 17243.

3.5.7.14 Service valves. Service valves shall be provided for the suction and discharge sides of the compressor and shall conform to MS 17248.

3.5.7.15 Indicating dial gages. The compound and pressure gages shall be provided conforming to MS 27900, except either 1/8 I.P.S. or 1/4 inch (6 mm) male flare connections may be used.

3.5.7.16 Dial-type thermometer distant reading. The dial-type distant reading thermometer shall be provided conforming to MS 27211. The thermometer shall be located on the instrument panel with a remote bulb secured within the hardening cabinet to indicate the temperature of the cabinet.

3.5.8 Controls. The controls provided for the plants shall automatically maintain the specified temperatures in 3.4.1 for hardening cabinet. The controls shall be provided to permit manual operation of the freezer head section.

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3.5.8.1 Disconnect switch. All power wiring shall terminate at an inclosed, fused, disconnect switch mounted within the freezer head compartment. The disconnect switch shall be located in a readily accessible location. A 20 foot (6 m) length of standard heavy duty 4-wire cable shall be furnished complete with a 4-wire male plug.

3.5.8.2 Motor starter switch (automatic). The motor control switch shall be type I, class 2 NEMA general purpose in drip-tight inclosure conforming to MIL-S-12514.

3.5.8.3 High and low pressure cutout control. A high and low pressure cutout control shall be provided conforming to MS 17843.

3.5.8.4 Manual switches. A push button manual starter with overloads shall be furnished for the control of the freezer head motor. A manually operated toggle switch shall be provided to control the solenoid valve. Both shall be marked with an "OFF-ON" position.

3.6 Government furnished property. The engine generator (model SF-5.0 MD/SIED) shall be furnished by the Government (see 6.6) for each type II unit. The engine generator unit assembly shall be mounted by the contractor within the structural framework as shown on drawings and secured to the base plate as required.

3.7 Dehydration and charging. The completely assembled refrigeration system shall be evacuated, dehydrated, and charged with the refrigerant specified in 3.3.10 and an operational quantity of oil specified in 3.3.11. The moisture indicator specified in 3.5.7.12 shall indicate dry when the refrigeration system is tested as specified in 4.4.2.

3.8 Solder-joints. All solder-joint connections, except the evaporator coil, condenser coil, and discharge line shall be made with the solder specified in 3.3.4. The evaporator coil, condenser coil, and discharge line solder connections shall be made with silver solder. Solder joint shall be properly cleaned before joining. Heavy solder beads of fillets shall be avoided and full fillet deposits shall be removed from the finished joint. After leak testing, all solder joints and tubing shall be sprayed with lacquer.

3.9 Finish. The interior and exterior surfaces of the inclosure, framework and base, guards, and grills, and auxiliary power unit shall be painted after first cleaning and preparing the surfaces with compound specified in 3.3.8, except where painting would be considered detrimental to the function of the ice cream plant. The priming paint and finish coat shall be as specified in 3.3.9.1 and 3.3.9.2. No paint shall be applied to the condenser fins or tubes. All burrs, rough edges, and slivers shall be removed.

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3.10 Electromagnetic compatability. Unless otherwise specified (see 6.2), the ice cream plants shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.7).

3.11 Lubrication.

3.11.1 Lubricants. The electric motors with accessories thereto and the condenser fan and fan shaft bearing shall be designed to operate under all test requirements and operating conditions when lubricated with the manufacturer's standard lubricants.

3.11.2 Lubrication fittings. Lubrication fittings shall be selected from MS 15005.

3.11.3 Initial service. The equipment shall be thoroughly lubricated before delivery to the Government and after such lubrication, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.11.4 Lubrication instructions. Unless otherwise specified (see 6.2), lubrication instructions shall be affixed in a convenient location for each major component or section of ice cream plant.

3.11.5 Lubrication precaution marking. Where the use of high pressure lubricating equipment (1,000 psi (6894 kPa) or greater) might damage grease seals or other parts, a warning shall be affixed to the equipment in a conspicuous location.

3.12 Marking.

3.12.1 Marking for identification. All equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130, except where otherwise specified herein. A metal name plate shall be attached by screws or rivets on the ice cream plant as shown on drawing. The metal name plate conforming to type III, composition A, class 1 or 2 of MIL-P-514. The metal name plate shall be marked with letters "U.S." in characters 2 inches (51 mm) high. In addition the nomenclature of item, capacity, type refrigerant, contract number, manufacturer's serial number, type number, and trade mark of such known characters as to be identifiable with the manufacturer's name and address.

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3.12.2 Instruction plate. Each ice cream plant, when assembled, shall have attached, as shown on drawing, a noncorrodible metal instruction plate. Instruction plate shall have etched on it the schematic wiring diagram and refrigeration cycle diagram of the unit. Plate(s) shall also contain instructions for "installation", "start-up", "operating", "shutdown", and preventive maintenance procedures. The instruction plates shall be 0.032 inch (0.8 mm) thick.

3.13 Workmanship. Sheet metal or structural sections used in fabrication of the ice cream plants shall not be kinked or bent. Bolts, nuts, and screws shall be drawn tight and so secured that they will not work loose under vibration. The surfaces of parts to be welded shall be clean and free of scale or other foreign matter. Welds shall be equal to the strength of sections joined. All scale or flux deposits shall be removed from the finished welds. All exposed welds at joints or seams in the final assembly shall be ground smooth and polished so that adjacent surfaces shall match. Adhesives, of a type and in the quantities recommended by the manufacturer for the purpose intended, shall be used where applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article examination. When a first article is required, it shall be examined for defects listed in table I, for compliance with dimensions specified, and tested as specified in 4.4.1 through 4.4.7. Failure to pass any of the tests specified, or the presence of any defect listed in table I, or any dimension not within specified requirements shall be cause for rejection of the first article.

4.3 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.3.2 In-process examination. Process examination shall be made of the following processes to establish that no deviations are made from the specified requirements. Whenever nonconformance is noted, corrections shall be made to the process and items affected:

(a) Wood preservation. Examination shall be made to determine conformance with method and application of preservative as specified in 3.5.

(b) Insulation installation. Examination shall be made to determine that insulation is installed as specified in 3.5.1 and applicable drawings.

(c) Welding and soldering. Examination shall be made to determine that welding and soldering is in conformance to 3.13 and 3.8 and applicable drawings.

(d) Metal surface cleaning and priming. Examination shall be made to determine that cleaning and priming is as specified in 3.3.8 and 3.3.9.1, respectively, and 3.9.

(e) Adhesive cementing. Examination shall be made to determine that adhesive cementing is applied as specified in 3.13.

(f) Tubing installation. Examination shall be made to determine that tubing installation is as specified in 3.5.

(g) Coating of electrical components with mildew-resistant varnish is applied on electrical components as specified in 3.5.

(h) Breaker strips are installed, sealed, and vulcanized as specified in 3.5.6.

4.3.2.1 Intermediate tests.

4.3.2.1.1 Hardening cabinet leak test. The hardening cabinet inner liner shall be leak tested as specified in 4.4.8. The sample unit shall be one completed liner ready for assembly. The inspection lot shall be the number of liners offered for inspection at one time. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

4.3.2.1.2 Density of insulation test. The density test for insulation shall be performed as specified in 4.4.9. The inspection lot shall be the amount of material used in one day's production. The sample unit for this test shall be a 6 inch (152 mm) cube of insulation.

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4.3.3 End product inspection. The lot shall be the number of ice cream plants of one type offered for inspection at one time. The sample unit for this examination shall be one completely fabricated portable ice cream plant of the specified type.

4.3.3.1 Visual examination. Examination of the end product shall be in accordance with the classification of defects set forth in table I. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified	X	
	Blistered, peeled, chipped, or area of no film		X
	Color of finish not as specified	X	
	Poor touch up, orange peel, color separation, smudge, fish scale, scratched, stained, over-spray, sag, imprint, foreign matter imbedded or finish is otherwise impaired affecting appearance		X
	Burrs, rough edges, and slivers	X	
	Paint on condenser fins or tubes	X	
Design	Panel must be removed for cranking (Type II only)		X
Construction and general workmanship (applicable to all components and assemblies)	Any part missing or not specified type	X	
Soldering (when applicable)	Missing, not adherent or incomplete	X	
	Heavy solder beads or fillets not removed		X
	Solder joints and tubing not sprayed with lacquer		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured or loose but not affecting operation		X
Assembly	Any component not properly assembled or secured e.g., tubing creased at bend restricting or partially restricting interior flow	X	
Marking identification	Missing, incomplete, improperly applied or illegible		X
Lubrication, instructions	Missing or illegible	X	
Lubrication precautionary marking	Missing, not as specified, not in proper location	X	
Instruction plate	Missing, not as specified, not in specified location, illegible, or incomplete	X	

4.3.3.2 Dimensional examination. Examination shall be made of ice cream plants to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end product. Each ice cream plant shall be tested in accordance with the tests specified in 4.4.2 through 4.4.5. Any nonconformance shall be cause for rejection of the unit. When a first article is not required, the initial production unit shall be tested as specified in 4.4.1, 4.4.6 and 4.4.7. Any nonconformance shall be cause for rejection of the unit.

4.3.4 Codes and standards compliance. Proof of compliance with 3.2.1 and 3.2.2 shall be made available to the Government representative.

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4.3.5 Inspection of packaging. An inspection shall be made to determine that preservation-packaging, packing, and marking as required by section 5 are complied with and are in accordance with MIL-R-12323 and MIL-G-52279, as applicable.

4.4 Tests.

4.4.1 Drop test. After complete assembly of all components of the ice cream plant and after the test specified in 4.4.5, the ice cream plant shall be placed on a level concrete floor and tilted 2 inches (51 mm) on each end and each side, one side and one end at a time, and free dropped four times on each end and each side to determine compliance with 3.5. Upon completion of this test, the ice cream plant shall be examined for any evidence of material or design failure. Any damage observed shall constitute failure of this test.

4.4.2 Moisture content of refrigerating system test. During and after the operation test of 4.4.5 and capacity test of 4.4.6, the combination moisture and liquid indicator shall be examined to determine compliance with 3.7. Moisture exceeding requirements shall constitute failure of this test.

4.4.3 Refrigerant leak test. To determine conformance to 3.5, all refrigerant containing parts shall be tested with a halide leak detector or other commercial method on completion of the tests specified in 4.4.1, 4.4.5, and 4.4.6. Any leaks shall constitute failure of this test.

4.4.4 Seal test. To determine conformance to 3.5, the proper seating and sealing of lid gaskets shall be tested at the four corners and midpoints on all lids by trying to insert a 0.005-inch (0.1 mm) leaf-gage between the gasket and the seating face. Observation shall be made that there is a resistance to the insertion of the leaf-gage. Inability of the lid gaskets to seat as specified shall constitute failure of this test.

4.4.5 Operation test. The completely assembled ice cream plant shall be operated for a period of 2 hours, and while the system is in operation, the following operations and observations shall be made on the freezer head section:

1. The freezer head refrigerant shut off valve shall be closed for this test and the freezer shall be filled with water. The mechanical parts of the freezer head section shall then be operated for 15 minutes after which time all seals including the shaft seal shall be observed for any leakage. Observation shall be made to determine all internal components function.

2. The refrigerating system shall continue to operate for the balance of the 2 hour period on the hardening cabinet only. During this period, the operation of the compressor, expansion valve, pressure control, and all other operating components shall be observed to be in adjustment and operating. Inability of the unit to operate for 2 hours shall constitute failure of this test.

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4.4.6 Capacity test. To determine compliance with 3.4.1, the following procedure shall be used: A completely assembled ice cream plant shall be placed in an ambient temperature of 110°F (43°C) for a minimum of 12 hours, with the lids of the hardening cabinet open, prior to test. The lids shall then be closed and the plant operated until the cabinet is down to temperature. The ambient temperature during this test shall be 110°F (43°C). As the freezer test is made the ice cream shall be packaged in 1/2-pint (0.002 m^3) containers and placed in the baskets in the hardening cabinet of the unit. The hardening cabinet temperature is to be the average of three air or product temperatures taken with thermocouples located no closer than 10 inches (254 mm) from each other or from sides, top or bottom of the cabinet liner. The temperature difference specified in 3.5.5 shall be the difference between the cabinet temperature and the pressure equivalent saturated temperature taken with a mercury column at the compressor suction valve. Total hardening time shall be taken from the time the last ice cream produced by the freezer is placed in the hardening cabinet. Inability of the unit to meet the capacity requirements of 3.4.1 shall constitute failure of this test.

4.4.7 Electromagnetic compatibility. When electromagnetic compatibility is required, the first article or initial unit of production, as applicable, shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and requirements meet MIL-STD-461.

4.4.8 Hardening cabinet inner liner leak test. To determine compliance with the requirements of 3.5, the inner liner of the hardening cabinet shall be placed in a jig or frame with stopper in place and filled with water. Observation shall be made that the stopper and all seams and joints do not leak. Any leakage shall constitute failure of this test.

4.4.9 Free rise density test for insulation. Prior to each day's production, one free foam sample of the insultation referenced in 3.3.5.2, a minimum 8 inch (203 mm) cube, shall be taken. From the center of the sample, a 6 inch (152 mm) cube shall be cut and weighed. The density shall be determined as follows:

$$\text{Weight of sample in pounds} \times 8 = \text{lbs/ft}^3$$
$$(\text{Weight of sample in kg/m}^3)$$

Inability of samples to conform to the density requirement of 3.3.5.2 shall constitute failure of the test.

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* 5. PACKAGING

* 5.1 Packaging requirements. The requirements for preservation-packaging, packing and marking for levels A, B, or C (commercial) shall be in accordance with MIL-R-12323 as specified (see 6.2). Each auxiliary power unit when supplied, shall be preserved-packaged, packed and marked in accordance with the level A, B, or C (commercial) requirements of MIL-G-52732.

6. NOTES

6.1 Intended use. The portable ice cream plants covered by this specification are intended for the production of ice cream for Military use in the field.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of unit required (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.3).
(d) If electrical characteristics for the ice cream plants are other than specified (see 3.4).
(e) Whether paint or color is other than specified (see 3.3.9.2).
(f) If electrical characteristics for fan are other than specified (see 3.5.7.6).
(g) Source and delivery of Government furnished property (see 3.6).
(h) When electromagnetic compatibility is not required (see 3.10).
(i) Whether lubrication instructions are waived (see 3.11.4).
(j) Selection of applicable levels of preservation and packaging, and packing (see 5.1).
(k) When drawings are required (see 6.5).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and number of units to be furnished. The contracting officer should include specific instructions regarding arrangements for inspection and approval of the first article.

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6.4 Engineering design conference. The following information should be included in the invitation for bids: Within 10 days after award of contract and prior to any work on the first article including ordering of materials, the contractor should request the contracting officer in writing to arrange an engineering design conference to discuss any questions concerning the details of Government drawings, instructions, and technical requirements of this specification. This conference is to assist the contractor in interpretation of the specification, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this specification nor relieve the contractor of his obligation to supply the equipment specified.

6.5 Drawings. When specified (see 6.2), reproducible record drawings of the end item shall be supplied with each contract by the contractor. These drawings shall include the assembly, subassemblies and component parts in sufficient detail to indicate construction methods, materials used, and dimensions.

6.6 Government furnished property. For each type II unit, the component listed below will be furnished to the contractor by the Government.

Item	Quantity	Nomenclature	Model
1	1	Generator, engine 5-Kilowatt, 208 volt, 60 Hz, 3 phase, 4 wire	SF-5.0 MD/SIED

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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